

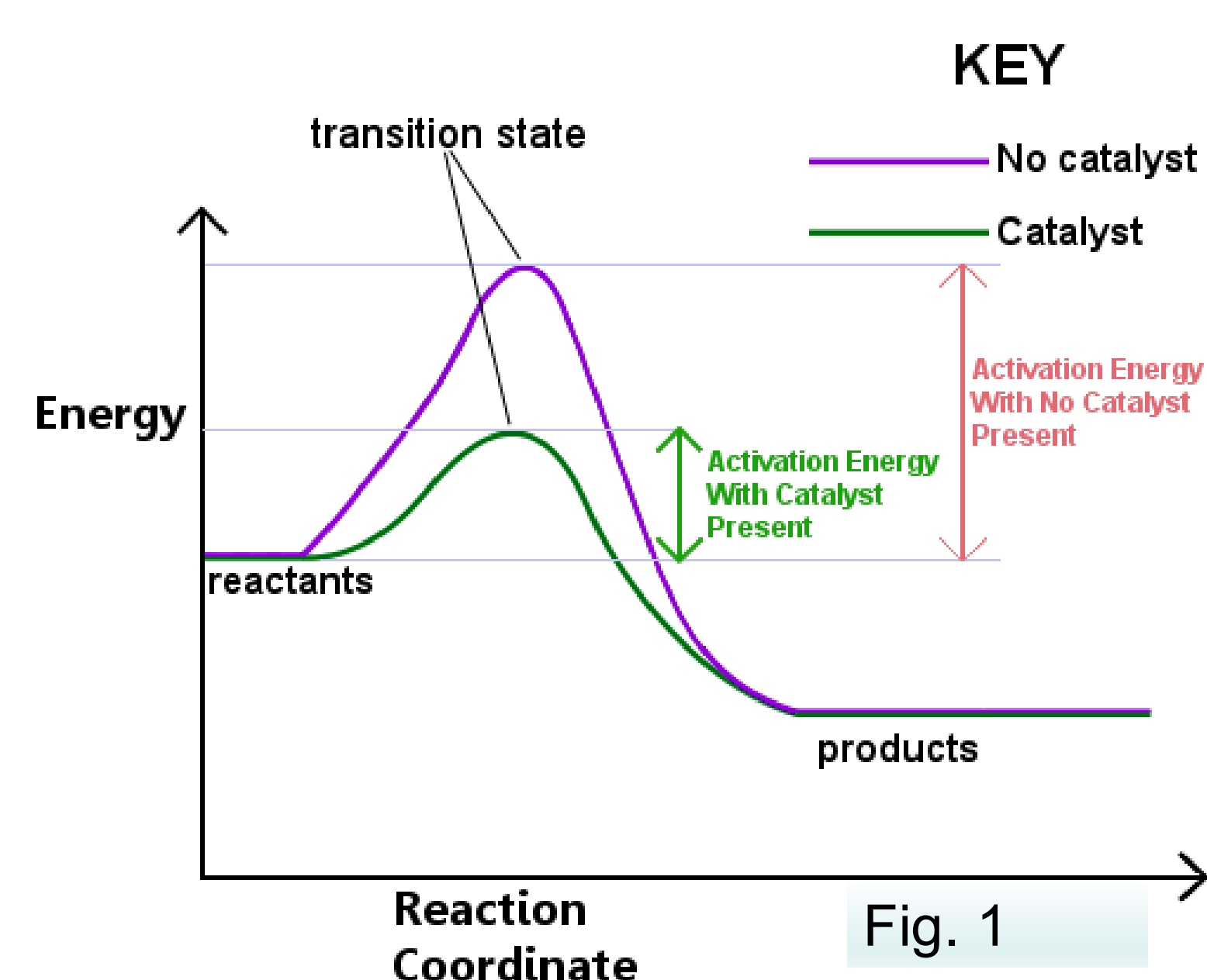
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Automating the Fragmentation of Proteins

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Background

Enzymes are biological catalyst that helps with the functions inside the body. They are composed of proteins, which in turn are composed of amino acids.



Goal

To predict the energy of an enzymatic reaction using AI/ML for better predictions.

Methodology

- Using the Nova system, provided by the Iowa State University, to run NWChem to obtain the energies of the amino acids.

Key Learnings

- Python language
- Basic Quantum Chemistry
- Basic Computational Chemistry
- Unix navigation
- Molecular modeling software

Next steps

- Gather all the energies of the amino acid.
- Calculate the energies of proteins.
- Simulate the energies of enzymes.

Intended Outcomes for the Summer

We aim to accurately simulate the energy of a single protein by combining the simulated energies of different amino acids.

Summer Research and Accomplishments

- Wrote a Python script for Protein Data Bank (PDB) files.
- Wrote a Python script to convert dipeptide XYZ files to NW files.
- Wrote a Python script to extract the energy of amino acids, the difference between the energies, and atomic charges.

ALA_ALA_0			
29			
0 1			
H	3.28300	6.00500	0.80700
C	3.26500	5.37700	1.69600
H	3.23800	4.32900	1.40000
C	4.51300	5.62300	2.51000
O	5.29000	4.70600	2.74000
N	4.70100	6.86200	2.96100
H	4.00500	7.55300	2.73600
C	5.86000	7.30700	3.74400
H	6.75400	6.87300	3.29300
C	5.96600	8.83400	3.60600
H	5.09600	9.31300	4.05700
H	6.86500	9.18600	4.11400
H	6.02900	9.11100	2.55300
C	5.83600	6.82400	5.21700
O	5.95100	7.62000	6.14800
N	5.66500	5.51700	5.43700
H	5.59700	4.92100	4.61900
C	5.52900	4.86800	6.74700
H	4.86200	5.48300	7.35300
C	4.83700	3.51400	6.52500
H	4.63700	3.04100	7.48800
H	3.89100	3.65800	6.00100
H	5.48100	2.85900	5.93700
C	6.85400	4.75300	7.54600
O	7.15900	3.72500	8.13900
N	7.68400	5.78700	7.56000
H	7.40700	6.64900	7.11100
H	8.54500	5.70500	

Fig. 3

```
start ALA_ALA_0
geometry
H      3.28300      6.00500      0.80700
C      3.26500      5.37700      1.69600
H      3.23800      4.32900      1.40000
C      2.38800      5.60700      2.29800
C      4.51300      5.62300      2.51000
O      5.29000      4.70600      2.74000
N      4.70100      6.86200      2.96100
H      4.00500      7.55300      2.73600
C      5.86000      7.30700      3.74400
H      6.75400      6.87300      3.29300
C      5.96600      8.83400      3.60600
H      5.09600      9.31300      4.05700
H      6.86500      9.18600      4.11400
H      6.02900      9.11100      2.55300
C      5.83600      6.82400      5.21700
O      5.95100      7.62000      6.14800
N      5.66500      5.51700      5.43700
H      5.59700      4.92100      4.61900
C      5.52900      4.86800      6.74700
H      4.86200      5.48300      7.35300
C      4.83700      3.51400      6.52500
H      4.63700      3.04100      7.48800
H      3.89100      3.65800      6.00100
H      5.48100      2.85900      5.93700
C      6.85400      4.75300      7.54600
O      7.15900      3.72500      8.13900
N      7.68400      5.78700      7.56000
H      7.40700      6.64900      7.11100
H      8.54500      5.70500      8.07400
end

basis
* library aug-cc-pvdz
end

echo

scf
direct
end

task scf energy
```

Fig. 2

Challenges

- Learning Python
- Many of the dipeptide files are unusable.

Example: L-asparagyl-L-asparagine (ASN ASN)

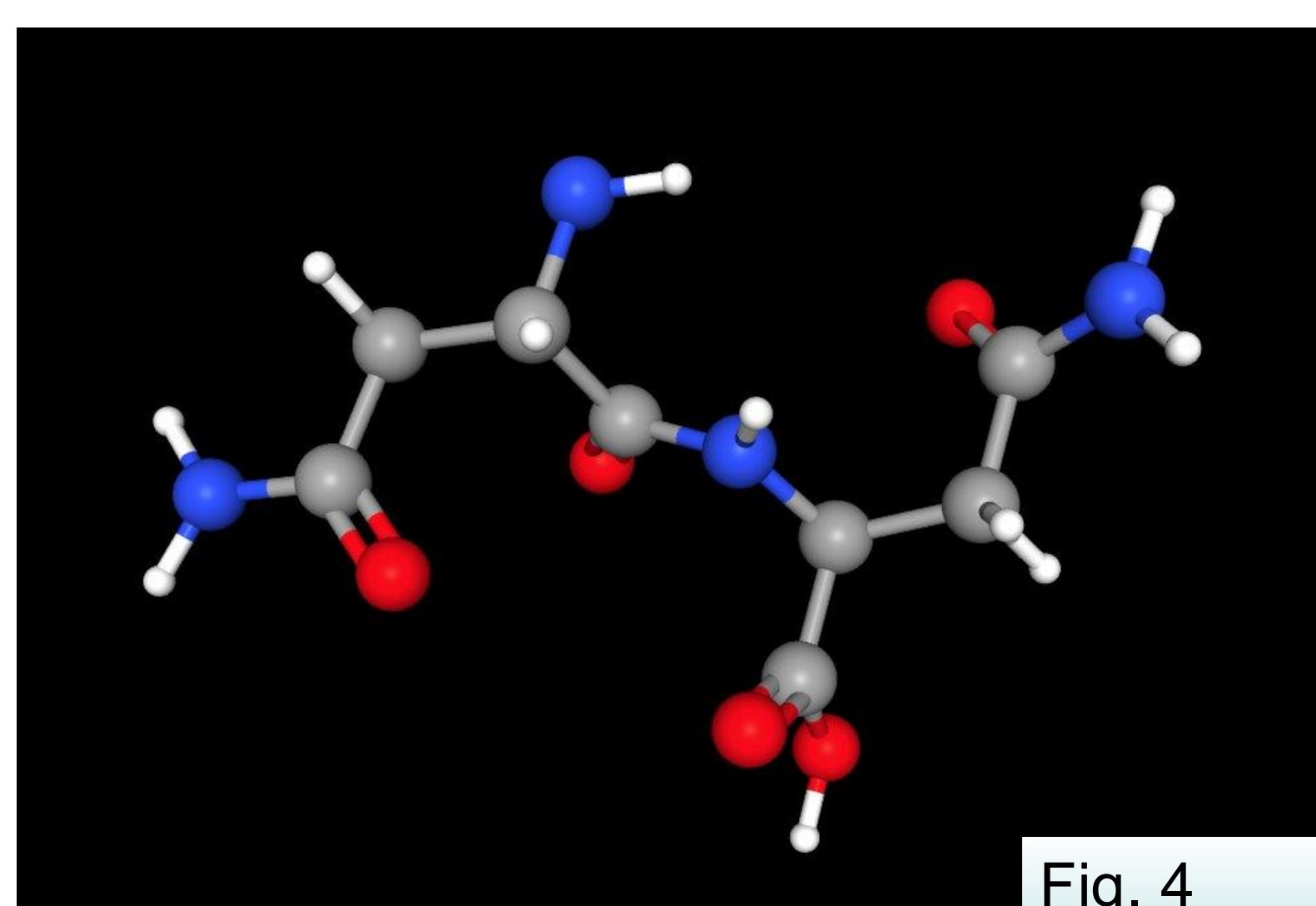


Fig. 4 was taken from PubChem.

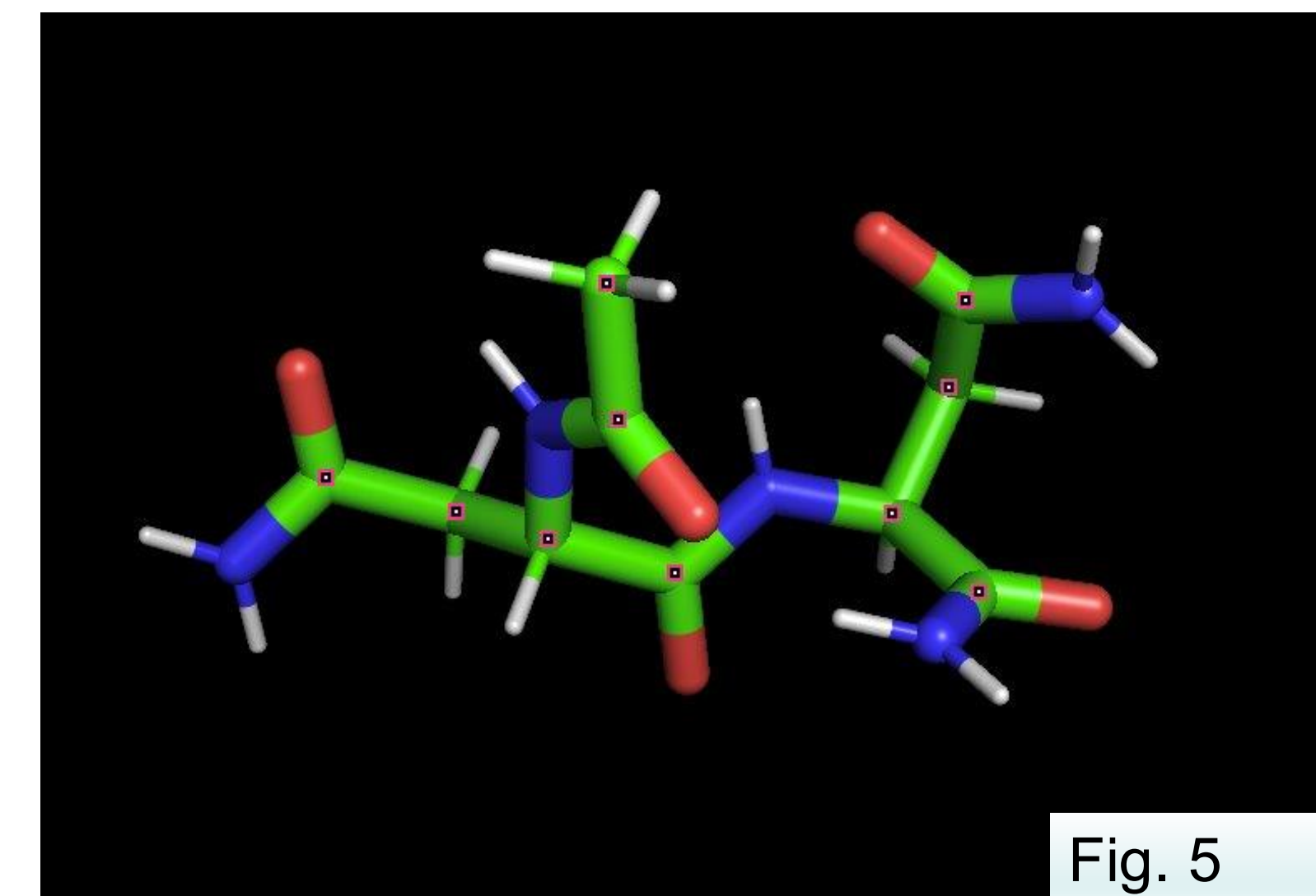


Fig. 5 was generated using PyMol.